

LEVEL DATA MANAGER SYSTEM

OVERVIEW

The Level Data Manager is a relatively simple system based on the script `LevelManager.cs`, which sits on the `LevelDataManager` game object. It's accessible through the singleton `LevelManager.LM`. It primarily acts as a container to track the current amount of carbon and money in the game. Long ago, in a galaxy far, far away, when Clean World was the *Untitled Carbon Capture Usage and Storage Simulation*, the Level Manager played a larger role, tracking stored carbon, the progression of years, and level states. But as it is, Clean World has become less of a simulation, so the Level Manager performs a much smaller number of functions.

HOW THE LEVEL MANAGER WORKS

Most of the Level Manager's work is done by the tiles themselves. Every time a tile is placed, it tells the level manager to increase the net carbon and money with the `AdjustNetMoney/AdjustNetCarbon` functions by the amount of carbon and money its tile scriptable object tells it to produce. And every time a tile is destroyed, it tells the level manager to reduce the net carbon and money using the same functions. (Note that activatable tiles do this when they're activated and deactivated, not when they're placed and destroyed.)

The Level Manager will increment and decrement the money and carbon amounts itself every time there's a carbon tick or a money tick, which each happen every four seconds, offset from each other by two seconds. The Level Manager also keeps track of when the carbon maxes out and prevents the carbon levels from going very far over the max.

INTERNAL API

These are some of the important public functions of the LevelManager class.

`void AdjustNetMoney(int _adjustNetMoney)`

Changes the current net money per tick by the input integer, “_adjustNetMoney”. Positive numbers increase it, negative numbers decrease it. This is called from tiles when they’re activated/deactivated or placed/destroyed.

`void AdjustNetCarbon(int _adjustNetCarbon)`

Changes the current net carbon per tick by the input integer, “_adjustNetCarbon”. Positive numbers increase it, negative numbers decrease it. This is called from tiles when they’re activated/deactivated or placed/destroyed.

`static bool overMaxCarbon()`

Returns true if the carbon is maxed out. Returns false if the total carbon isn’t maxed out.

`void AdjustMoney(int value)`

Changes the amount of money. Positive values increase it, and negative values decrease it. Used by the tiles when they’re purchased to reduce their price from the total amount of money that there is. Also used by the level manager every money tick, when buildings produce money.

`void AdjustCarbon(int value)`

Changes the amount of carbon. Called by the level manager itself every carbon tick.

`void ResetData()`

Sets all of the values back to their defaults.

Getters and Setters

Note: The Level Manager uses more getters and setters than the rest of the scripts. They tend to clutter code, so I try to avoid them, just using C#’s lovely auto property.

`int GetMoney()`

Returns how much money there currently is.

`void SetMoney(int _money)`

Sets how much money there is equal to the value of the input “_money” variable.

`int GetCarbon()`

Returns how much carbon there currently is.

`int GetMoney()`

Returns how much money there currently is.



```
int getMaxCarbon()
```

Returns what the current maximum amount of carbon allowed it.

```
int setMaxCarbon()
```

Sets how much carbon is allowed before the carbon meter maxes out and the building of carbon-producing structures is prohibited.

-Graydon Bruyn, May 2026